CHT Analysis of the NASA C3X Turbine Vane

Lois Kostner

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1 Objectives and scope

This report presents a conjugate heat transfer (CHT) simulation of the NASA C3X turbine vane. The C3X is an internally cooled guide vane with ten radial cooling holes. It was tested by NASA in a linear cascade, and the measured data are publicly available [3]. For this reason, it is a common test case for the validation of CFD models of cooled vanes.

In a CHT simulation the fluid and the solid are solved together. The wall temperature is therefore a result of the simulation, and not a boundary condition.

The aim of this work is to reproduce one operating point of the NASA experiment, run 157, with ANSYS Fluent, and to check how close the simulation is to the measurements. The boundary conditions and the material properties are taken from Norheim and Amzin [5], who simulated the same run. The mesh strategy and the options of the turbulence model follow Froissart and Ochrymiuk [2].

More specifically, the report aims to:

- build the geometry and a fully structured mesh of the vane, of the cooling channels and of the flow passage;
- set up the CHT simulation with the conditions of run 157;
- check the numerical quality of the solution;
- compare pressure, wall temperature and wall heat flux at midspan with the reference data, and discuss the differences.

The scope is limited to one simulation of one operating point. A mesh independence study is outside the scope of this work: only one mesh was used, with a size close to the one of a published study of the same vane [2]. Sensitivity studies on the turbulence model and on the boundary conditions are also not included.

This project was also a learning activity. The main new topics learned during the work are CHT modelling and structured meshing.

2 Reference test case

2.1 NASA C3X experiment

The experimental data come from the NASA report by Hylton et al. [3]. The tests were carried out in a linear cascade of three vanes, and only the central vane was instrumented. The vane has a constant cross-section along the span, so the flow at midspan is close to two-dimensional. It is made of ASTM 310 stainless steel and is cooled by ten radial holes, each fed by a separate line. All the experimental data used in this work come from this report.

The same profile was also used in two later NASA campaigns: one with film cooling at the leading edge [6] and one with high inlet turbulence [1]. These two reports are used here only for the vane coordinates (Appendix B) and, in the second case, for the discussion of the boundary-layer transition.

The static pressure and the surface temperature were measured near midspan, together with the mass flow rate and the temperature of the coolant in each hole. The external heat transfer coefficient was not measured directly. It was computed from a two-dimensional finite element solution of the heat conduction in the vane section.

The uncertainties declared by NASA are ±1 K on the surface temperature, ±11 K on the gas temperature and ±0.7 kPa on the pressures. The uncertainty on the heat transfer coefficient goes from ±6% in the front part of the vane to ±24% near the trailing edge.

The simulated case is run 157 of [3], identified by the code 4521. Its main conditions are:

- inlet total pressure of about 413 kPa and inlet total temperature of 818 K;
- inlet Mach number of 0.17 and exit Mach number of 0.89;
- exit Reynolds number of $2.44 \cdot 10^6$, based on the true chord;
- inlet turbulence intensity of 8.3%.

2.2 Reference numerical studies

Norheim and Amzin [5] simulated run 157 with Simcenter STAR-CCM+ and the k–ω SST model, on a polyhedral mesh of 1.9 million cells. This paper is the source of the domain, of the boundary conditions and of the material properties used here. Its results are also used for comparison.

Froissart and Ochrymiuk [2] used ANSYS ICEM CFD and ANSYS Fluent, as in this work, with a fully structured mesh of 13.5 million elements. They studied the effect of the options of the k–ω SST model. They simulated run 158, which has a lower pressure and a lower Reynolds number, so their curves cannot be compared directly with the present ones. From this paper, the present work takes the mesh strategy and the model options.

3 Geometry

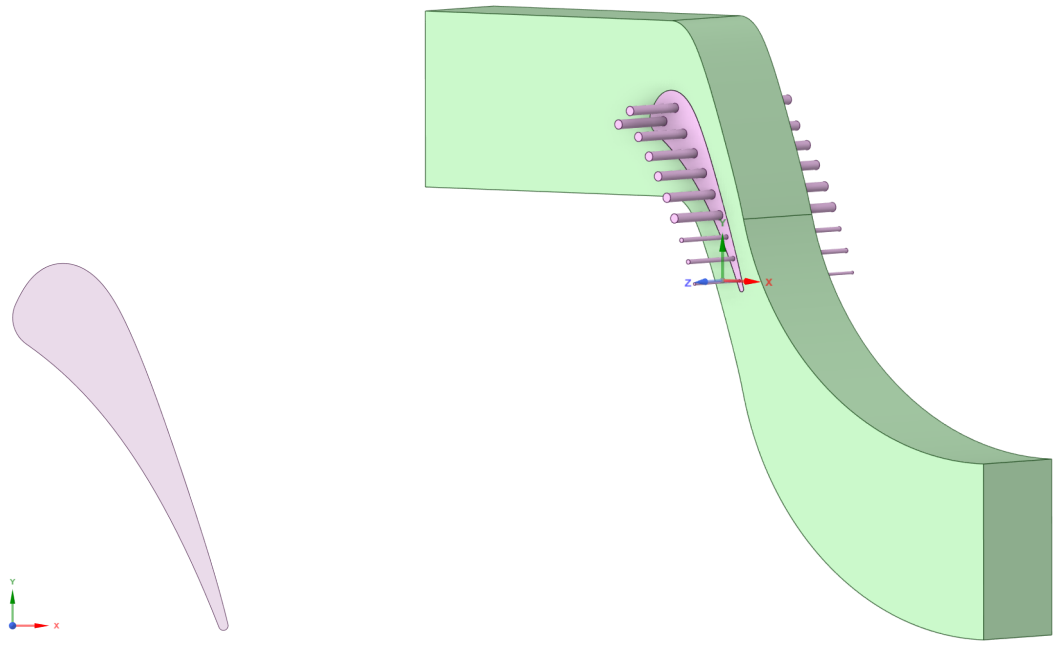
Following the NASA setup, the cascade is modelled as an infinite row of identical vanes. The flow is therefore periodic, with a period equal to the vane spacing.

The 2D profile was obtained by interpolating the NASA coordinates [3], which are listed in Tab. 12 of Appendix B. One coordinate of the original table contains a typing error and was corrected, as explained in the same appendix. The resulting profile is shown in Fig. 1a.

The vane has an axial chord of 78.16 mm, a spacing of 117.73 mm and a height of 76.2 mm [3]. Holes 1 to 7 have a diameter of 6.3 mm, holes 8 and 9 of 3.1 mm, and hole 10 of 1.98 mm [5]. The smallest holes are in the thin rear part of the vane.

The 3D geometry was built in ANSYS SpaceClaim and reproduces the domain of Norheim and Amzin [5]. It is shown in Fig. 1b. The fluid domain has a height equal to one vane spacing. In this way the upper and lower boundaries have the same shape and can be coupled as periodic boundaries. The inlet is 187.2 mm upstream of the leading edge, and the outlet is 200 mm downstream of the trailing edge.

The flow around the vane is assumed to be two-dimensional. However, the domain must be three-dimensional, because the cooling channels cross the vane along the span. In addition, the coolant becomes warmer along the channels, so the metal temperature changes along the span. The channels are extended by 50 mm on both sides of the vane, so that the coolant flow can develop before it enters the vane.



(a) Vane profile.

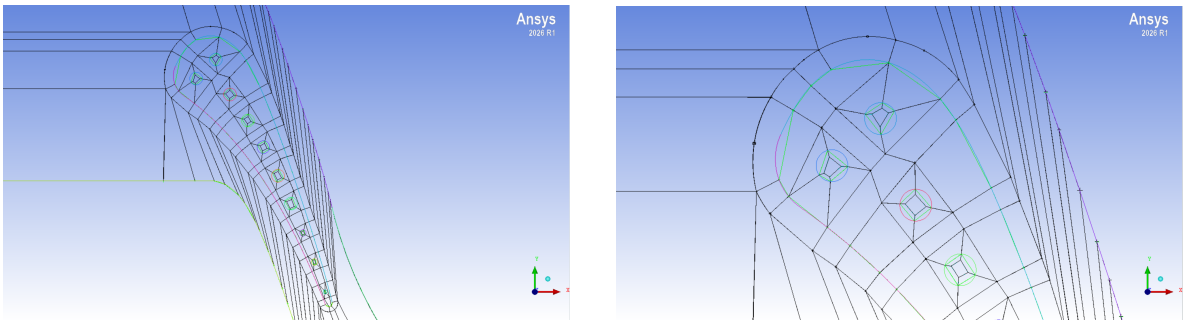
(b) Computational domain.

Figure 1: C3X vane geometry.

4 Mesh

4.1 Blocking strategy

The mesh is fully structured and was generated in ANSYS ICEM CFD with the blocking method: the domain is divided into blocks, and each block is filled with hexahedral cells. The blocking is shown in Fig. 2.



(a) Full vane. (b) Half vane.

Figure 2: O-grid blocking around the C3X vane.

An O-grid is placed around the vane. Fig. 2a shows the whole vane, and Fig. 2b shows a closer view of half of it. In an O-grid the blocks are wrapped around the body. Very thin cells can therefore be used near the wall without refining the rest of the domain. The same idea is used inside each cooling channel.

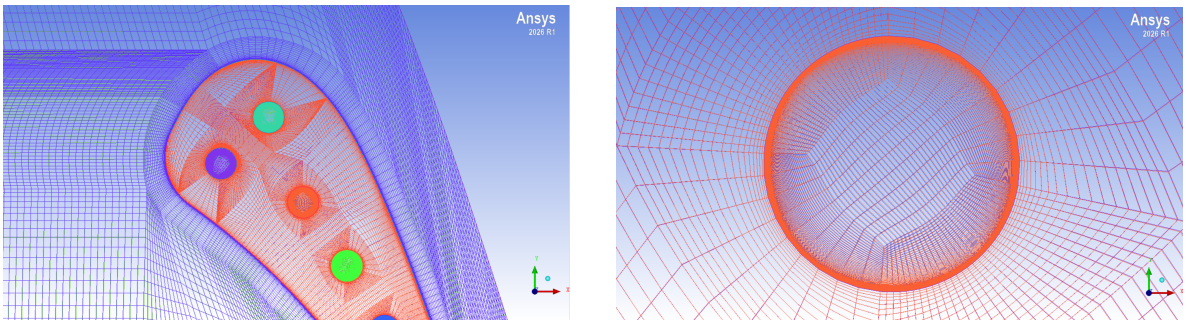
The mesh is conformal at the interfaces between fluid and solid. The nodes on the two sides are in the same positions, so the heat flux is exchanged without interpolation.

4.2 Mesh size and near-wall resolution

The mesh has about 11 million cells, close to the 13.5 million elements of Froissart and Ochrymiuk [2]. Both reference studies found their results to be almost independent of the mesh at much smaller sizes: above 1.4 million polyhedral cells in [5], and above 3.5 million hexahedral elements in [2].

These results are a useful indication for the mesh resolution, but they cannot be transferred automatically to the present simulation, because the meshes, the solvers and the setups are different. In addition, both studies checked mesh independence on a single temperature value: the maximum vane temperature in [5], and the leading-edge temperature in [2]. This does not guarantee that every local value of the heat flux is independent of the mesh. A mesh independence study was not carried out in this work, so the numerical error of the present mesh is not known.

The height of the first cell is about 1 μm, chosen to obtain $y^+ \approx 1$ on the vane surface and inside the holes. With this resolution the turbulence model is solved down to the wall, without wall functions. This is essential for the heat flux, which depends on the temperature gradient at the wall. The mesh is shown in Fig. 3.



(a) Half vane. (b) Cooling channel.

Figure 3: Structured mesh.

5 Solver setup

5.1 Models and numerical methods

The simulation was run in ANSYS Fluent with the steady, pressure-based solver and with the energy equation active. Turbulence is modelled with the k–ω SST model [4], as in both reference studies. The flow is treated as fully turbulent: no transition model is used.

Following [2], viscous heating, curvature correction, Kato–Launder production and production limiter are all active. According to the same authors, these options can change the wall temperature by 7 to 40 K, with the largest effect from viscous heating. The operating pressure is set to zero, so all the pressures given below are absolute.

The pressure–velocity coupling uses the SIMPLEC scheme, with least-squares gradients and second-order upwind schemes for all the equations. The under-relaxation factors were kept at their default values. Only the pressure factor was reduced, from 0.3 to 0.25, to obtain a stable convergence.

5.2 Material properties

The hot gas and the coolant are both air, with the properties of Tab. 1. The specific heat and the thermal conductivity are polynomials of the temperature, $\phi(T) = \sum_i a_i T^i$, with the coefficients listed in order of increasing power.

Table 1: Air properties.

Property	Method	Value / coefficients
Density ρ	ideal-gas	ideal-gas law
Specific heat c_p	polynomial	$a_0 = 957.110256$ $a_1 = 2.365234 \cdot 10^{-1}$ $a_2 = 5.141114 \cdot 10^{-6}$ $a_3 = -3.3917446 \cdot 10^{-9}$ $a_4 = -6.0929646 \cdot 10^{-12}$
Thermal conductivity λ	polynomial	$a_0 = -9.157270 \cdot 10^{-4}$ $a_1 = 9.524484 \cdot 10^{-5}$ $a_2 = -5.049820 \cdot 10^{-8}$ $a_3 = 1.545865 \cdot 10^{-11}$
Viscosity μ	sutherland	$\mu_0 = 1.7894 \cdot 10^{-5}$ kg/(m·s) $T_0 = 273.11$ K $S = 110.56$ K
Molar mass M	constant	28.966 kg/kmol

The air properties come from [5]. The only difference is the thermal conductivity. The paper uses a Sutherland-type law, which is not available in Fluent for this property, so it was replaced by a polynomial fitted on the same law.

The vane is made of ASTM 310 stainless steel, with the properties of Tab. 2, also taken from [5]. The thermal conductivity increases linearly with the temperature.

Table 2: Steel properties (ASTM 310).

Property	Method	Value / coefficients
Density ρ	constant	8030 kg/m^3
Specific heat c_p	constant	502 J/(kg·K)
Thermal conductivity λ	polynomial	$a_0 = 9.9105$ $a_1 = 1.15 \cdot 10^{-2}$

Froissart and Ochrymiuk use a conductivity that is 12% to 17% higher for the same steel [2]. This property is therefore one of the uncertain inputs of the problem.

5.3 Boundary conditions

The inlet of the main flow is a pressure inlet with the conditions of Tab. 3. The pressure p_{init} is used only to initialise the solution. The outlet is a pressure outlet with the conditions of Tab. 4, where the subscript bf indicates the backflow values. Both tables correspond to run 157 [5].

Table 3: Main inlet conditions.

p_0 [Pa]	p_{init} [Pa]	T_0 [K]	I [%]	μ_t/μ [–]
413 286	405 033	818	8.3	30

Table 4: Main outlet conditions.

p [Pa]	$T_{0,\mathrm{bf}}$ [K]	I_{bf} [%]	$(\mu_t/\mu)_{\mathrm{bf}}$ [–]
254 172	818	8.3	30

The inlets of the ten cooling channels are mass-flow inlets with the conditions of Tab. 5, taken from [5]. The total coolant mass flow rate is 0.17027 kg/s.

Table 5: Cooling channel inlet conditions.

Zone	$\dot{m}$ [kg/s]	T_0 [K]	D_h [m]	I [%]
`inlet_channel_1`	0.02220	342	0.00630	10
`inlet_channel_2`	0.02210	344	0.00630	10
`inlet_channel_3`	0.02180	335	0.00630	10
`inlet_channel_4`	0.02280	336	0.00630	10
`inlet_channel_5`	0.02250	330	0.00630	10
`inlet_channel_6`	0.02250	355	0.00630	10
`inlet_channel_7`	0.02160	336	0.00630	10
`inlet_channel_8`	0.00744	350	0.00310	10
`inlet_channel_9`	0.00477	377	0.00310	10
`inlet_channel_10`	0.00256	387	0.00198	10

The mass flow rates are the values measured by NASA [3]. The inlet temperatures, instead, are not reported by NASA, which gives only an average temperature for each hole at the measurement plane. The values of Tab. 5 are the ones adopted by Norheim and Amzin. They are 10 to 16 K lower than the NASA averages in the seven large holes, and 26 to 60 K lower in the three small holes. Froissart and Ochrymiuk make a different choice and assume 293.15 K for all the holes [2]. The coolant inlet temperature is therefore a second uncertain input.

The outlets of the channels are pressure outlets. Their pressure is discussed in Section 5.4.

The upper and lower boundaries of the main flow are periodic, and the hub and shroud planes are adiabatic, as in [5]. The vane surface and the walls of the channels are coupled walls. No temperature and no heat flux are imposed on them: the solver computes both, by requiring that they are the same on the fluid side and on the solid side.

5.4 Channel outlet pressure

The outlet pressure of the cooling channels is not given by any of the sources, so it had to be chosen. It is therefore a modelling assumption of this work, and it does not necessarily reproduce the experimental conditions. The only constraint used for the choice is that the channels must not be choked.

The mass flow rate through a channel of area A can be written as

$$\dot{m} = \rho\, u\, A = p\, A\, M \sqrt{\frac{\gamma}{R\,T_0}} \sqrt{1 + \frac{\gamma - 1}{2} M^2}, \tag{1}$$

where p is the static pressure, M the Mach number, T_0 the total temperature, γ the ratio of specific heats and R the gas constant. For an imposed mass flow rate, the Mach number increases when the pressure decreases. The flow becomes sonic ($M = 1$) when the pressure reaches the critical value

$$p^* = \frac{\dot{m}}{A} \sqrt{\frac{R\,T_0}{\gamma}} \sqrt{\frac{2}{\gamma + 1}}. \tag{2}$$

If the outlet pressure is lower than p^*, the channel is choked: the flow is sonic at the outlet, and the imposed pressure can no longer be respected. The outlet pressure must therefore satisfy

$$p_{\text{outlet}} > p^*. \tag{3}$$

Tab. 6 reports p^* for the ten channels, computed with $\gamma = 1.4$, $R = 287$ J/(kg·K) and the inlet values of Tab. 5. The most critical channel is channel 8, with $p^* \approx 2.4 \cdot 10^5$ Pa.

Table 6: Critical pressure of the cooling channels.

Channel	$\dot{m}/A$ [kg/(m^2·s)]	p^* [kPa]	p_{outlet}/p^* [–]	M_{outlet} [–]
1	712.2	172	8.7	0.126
2	709.0	172	8.7	0.125
3	699.3	167	9.0	0.122
4	731.4	175	8.6	0.128
5	721.8	171	8.8	0.125
6	721.8	178	8.4	0.130
7	692.9	166	9.0	0.121
8	985.7	241	6.2	0.176
9	632.0	160	9.4	0.117
10	831.4	214	7.0	0.156

The pressure imposed at the ten outlets is

$$p_{\text{outlet}} = 1.5 \cdot 10^6 \text{ Pa}, \tag{4}$$

which is at least six times higher than p^* in every channel. The table also reports the outlet Mach number M_{outlet}, obtained from Eq. (1) with $p = p_{\text{outlet}}$. It is below 0.2 in all the channels. Under this assumption, therefore, the coolant flow remains subsonic and far from choking. The coolant becomes warmer along the channel, which increases p^*, but this effect is small compared with the margin.

Two more remarks follow from this result. First, the paper gives the coolant inlet temperatures as static values, while the mass-flow inlet of Fluent requires total values. With these low Mach numbers the difference is only 1 to 2 K, so the values of the paper were used directly.

Second, for a fixed mass flow rate and temperature, the Reynolds number of an ideal gas in the channels does not depend on the pressure level. This suggests that the choice has a limited effect on the heat transfer in the holes, but it does not prove it: the density, the velocity and the Mach number of the coolant change with the pressure. No sensitivity analysis on the outlet pressure was performed, so the dependence of the thermal solution on this assumption is not known.

6 Post-simulation checks

6.1 Residuals

The residuals reached at the end of the run are listed in Tab. 7.

Table 7: Residuals at the end of the run.

Residual	Value
`continuity`	$< 1 \cdot 10^{-1}$
`x-velocity`	$< 4 \cdot 10^{-5}$
`y-velocity`	$< 3 \cdot 10^{-5}$
`z-velocity`	$< 3 \cdot 10^{-6}$
`energy`	$< 2 \cdot 10^{-5}$
`k`	$< 8 \cdot 10^{-3}$
`omega`	$< 4 \cdot 10^{-4}$

The velocity and energy residuals are low, while the continuity residual stayed rather high. Froissart and Ochrymiuk also report continuity as the most difficult residual [2]. In Fluent this residual is scaled with its largest value in the first five iterations, so it is not an absolute measure of the mass error. For this reason, the conservation of mass and of energy was checked directly.

6.2 Mass and energy balances

The mass and energy imbalances were computed as:

$$\varepsilon_{\dot{m}} = \left| \frac{\dot{m}_{\text{inlet}} + \dot{m}_{\text{outlet}}}{\dot{m}_{\text{inlet}}} \right|, \qquad \varepsilon_{\dot{q}} = \left| \frac{\dot{q}_{\text{inlet}} + \dot{q}_{\text{outlet}} + \dot{q}_{\text{walls}}}{\dot{q}_{\text{inlet}}} \right| \tag{5}$$

The quantities that leave the zone are negative, and $\dot{q}_{\text{walls}}$ is the heat exchanged with the vane through the coupled walls. The values for the main flow and for the ten cooling channels are reported in Tab. 8.

Table 8: Mass and energy imbalances.

Zone	$\varepsilon_{\dot{m}}$ [%]	$\varepsilon_{\dot{q}}$ [%]
Main flow	0.019412	0.008920
Channel 1	0.000295	0.004524
Channel 2	0.001495	0.016306
Channel 3	0.000361	0.005782
Channel 4	0.000296	0.004963
Channel 5	0.000318	0.006858
Channel 6	0.000582	0.002579
Channel 7	0.001157	0.005386
Channel 8	0.000515	0.001357
Channel 9	0.000234	0.007991
Channel 10	0.000286	0.011026

All the imbalances are below 0.02%. Mass and energy are therefore well conserved in every fluid zone, even if the continuity residual is high. Among the channels, the largest values are found in channel 2.

For the main flow, the energy error is divided by the inlet energy flow rate, which is much larger than the heat exchanged with the vane. This check is therefore less strict for the main flow than for the channels.

6.3 Near-wall resolution

The value of y^+ was checked on the vane surface and on the walls of the holes. It is below 1.5 everywhere, so the first cell is inside the viscous sublayer, as required when the turbulence model is solved down to the wall.

7 Results

7.1 Reference data

All the results are taken on the vane surface at midspan, where the NASA instrumentation is located. The position is given by X/L, where X is the axial distance from the leading edge and $L = 78.16$ mm is the axial chord. This coordinate is negative on the pressure side and positive on the suction side. As in [5], the pressure is divided by 413 286 Pa, the temperature by 811 K and the heat flux by 10^5 W/m^2. The value of 811 K is the nominal gas temperature of the NASA tests, while the actual gas temperature of run 157 is 818 K.

Each figure shows the present simulation, the NASA measurements of run 157 (27 pressure taps and 73 thermocouples) [3] and the simulation of Norheim and Amzin [5]. Their curves were digitised from the figures of the paper, so they contain a small reading error.

NASA does not report the heat flux, but the heat transfer coefficient h. The experimental heat flux was therefore reconstructed from h and from the wall temperature T_w as

$$q_{\mathrm{NASA}} = h\,(T_g - T_w), \qquad T_g = 818 \text{ K}. \tag{6}$$

This expression follows the NASA definition of h: the wall heat flux of the conduction solution divided by the difference between the gas temperature and the surface temperature [3]. The gas temperature of run 157 is 818 K.

The reconstructed heat flux is therefore not an independent measurement. It depends on the NASA data reduction, and it is not independent of the temperature data. Its accuracy depends on two more points. The first is the gas temperature used in the definition of h, which is assumed here to be 818 K: with the nominal value of 811 K, the reconstructed heat flux would be about 3% to 4% lower. The second is the uncertainty of h, which changes along the vane, from ±6% in the front part to ±24% near the trailing edge.

7.2 Error measures

To measure the agreement, the simulated curve is interpolated at the positions of the N NASA stations. At station i, the absolute error of a quantity ϕ is

$$e_{\phi,i} = \left|\phi_{\mathrm{CFD},i} - \phi_{\mathrm{NASA},i}\right|, \tag{7}$$

where ϕ is the pressure p, the wall temperature T or the heat flux q. Three values are computed from it: the mean absolute error, the mean relative error and the maximum error.

$$\bar{e}_\phi = \frac{1}{N}\sum_{i=1}^{N} e_{\phi,i}, \qquad \bar{e}_\phi = \frac{1}{N}\sum_{i=1}^{N} \frac{e_{\phi,i}}{\phi_{\mathrm{NASA},i}}, \qquad e_{\phi,\max} = \max_i e_{\phi,i} \tag{8}$$

The two mean errors are indicated with the same symbol and are distinguished by their units: the mean relative error is given in percent. For the pressure, for example, the errors are $\bar{e}_p$ and $e_{p,\max}$.

The errors are computed on all the NASA stations: 27 for the pressure, and 73 for the temperature and the heat flux. The same procedure is applied to the digitised curves of Norheim and Amzin, so that the two simulations are evaluated in the same way. The errors published in [5] are not used in the tables, because the paper does not define how they are computed.

7.3 Pressure distribution

Fig. 4 shows the normalised static pressure at midspan, and Tab. 9 reports the errors.

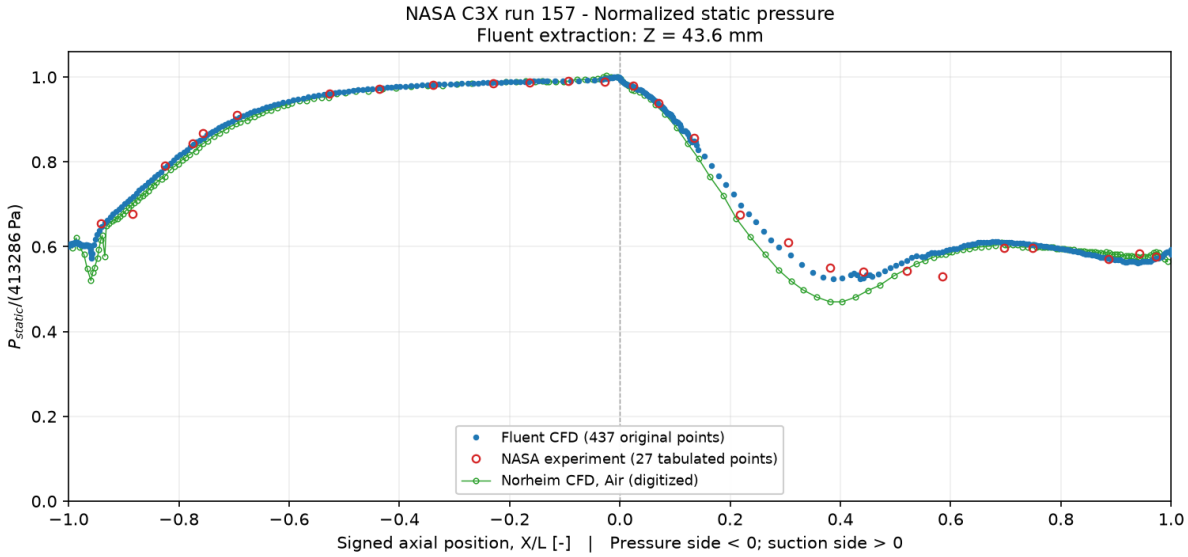


Figure 4: Normalised static pressure at midspan.

On the pressure side, the pressure stays close to the inlet total pressure over the first half of the chord, and then it decreases towards the trailing edge. On the suction side the flow accelerates strongly. The pressure reaches its minimum at $X/L \approx 0.44$, where the Mach number is about 1. After the minimum the pressure rises slightly and then changes little.

Table 9: Pressure errors with respect to the NASA data.

Simulation	$\bar{e}_p$ [kPa]	$\bar{e}_p$ [%]	$e_{p,\max}$ [kPa]
Present work	5.04	1.95	23.30
Norheim and Amzin	8.86	3.30	35.50

The agreement of the present simulation with the experiment is good: the mean relative error is below 2%. The error is larger on the suction side than on the pressure side, where the simulated curve passes through almost all the experimental points.

The maximum error $e_{p,\max}$ is on the suction side, at $X/L = 0.58$. In the experiment the low-pressure region is flatter and longer, while in the simulation the minimum is sharper and the pressure starts to rise earlier. This maximum is about 10% of the local pressure, so a small mean error does not exclude larger local errors.

With the same procedure, the digitised curve of the reference simulation gives larger errors. On the suction side it is below the experiment, which means a stronger acceleration: around the pressure minimum it reaches a Mach number of about 1.1, and its maximum error is in this region. The authors themselves report that their model overpredicts the acceleration on the suction side [5].

However, the mean relative error obtained here for the reference simulation, 3.30%, does not agree with the 1.39% published by its authors. The paper does not define how its average error is computed, or at which points. The difference was examined, but it could not be reconciled with the available information. The reading error of the digitised curve is too small to explain it. A different definition of the average could explain it: for example, an average in which positive and negative differences cancel gives between 1.5% and 1.6% with the digitised curve. This remains a hypothesis.

For this reason, Tab. 9 is not used to state that the present simulation is more accurate than the reference one. It only shows that, with the procedure used here, the two simulations have errors of the same order.

Froissart and Ochrymiuk found that the pressure distribution is almost independent of the turbulence model and of its options [2]. A possible reason for the difference between the two curves is therefore the different mesh. This is only a hypothesis, because no mesh study was carried out in this work.

7.4 Wall temperature distribution

Fig. 5 shows the normalised wall temperature at midspan, and Tab. 10 reports the errors.

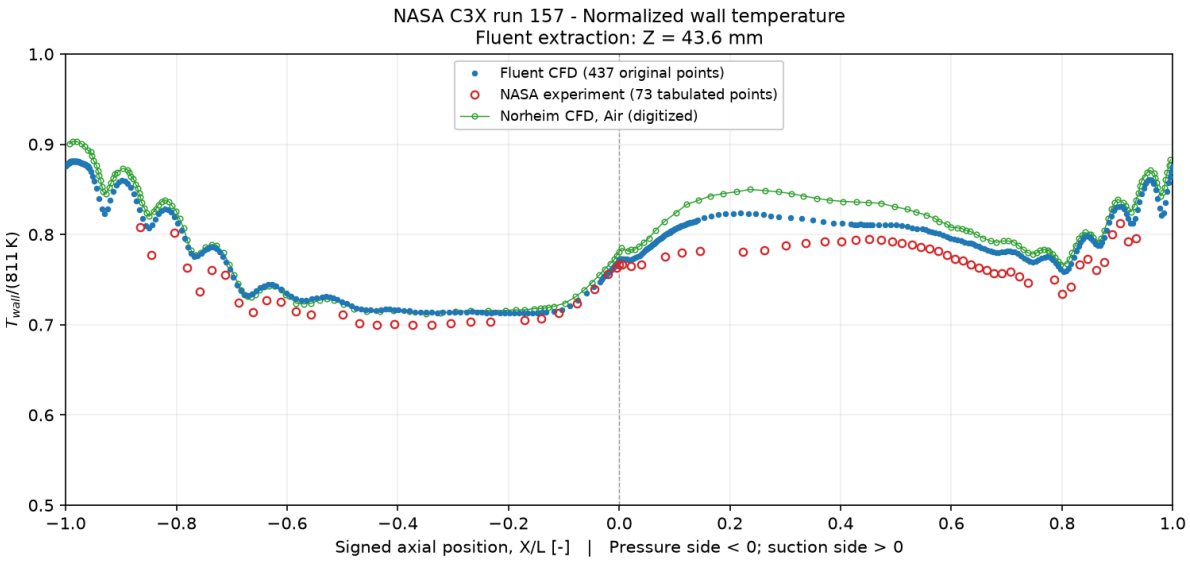


Figure 5: Normalised wall temperature at midspan.

The simulation reproduces the shape of the measured distribution. The coldest region is the front half of the pressure side, at about 580 K. The suction side is warmer, at about 660 K. The hottest points are close to the trailing edge, at about 715 K, where the vane is thin and is cooled only by the smallest holes. In the rear part the curve is wavy: each local minimum corresponds to a cooling hole close to the surface.

Table 10: Wall temperature errors with respect to the NASA data.

Simulation	$\bar{e}_T$ [K]	$\bar{e}_T$ [%]	$e_{T,\max}$ [K]
Present work	15.7	2.55	34.6
Norheim and Amzin	23.8	3.82	54.8

The simulation is hotter than the experiment at every station. The error is therefore not a random scatter, but a general offset, and $\bar{e}_T$ is its mean value. The offset is small at the leading edge, and it is larger on the suction side than on the pressure side. The NASA uncertainty on the wall temperature is only ± 1 K, so the offset cannot be explained by the thermocouples.

The maximum error $e_{T,\max}$ is on the front part of the suction side, at $X/L = 0.22$, and it is about twice the mean error. In this region a local error is therefore added to the general offset.

With the same procedure, the reference simulation shows the same behaviour, with larger errors. The two curves are close on the pressure side, while on the suction side the present one is closer to the experiment. Both have their maximum error at the same station. For the temperature, the mean

relative error computed here for the reference simulation is very close to the 3.78% reported in the paper [5], so in this case the two evaluations are consistent.

Froissart and Ochrymiuk also find their maximum error, about 20 K, in the most curved part of the suction side [2]. Three simulations with the same turbulence model therefore have their largest error in the same region. This suggests that the local error is related to the turbulence model, but it does not prove it. A possible explanation is discussed in Section 7.5.

7.5 Wall heat flux distribution

Fig. 6 shows the normalised wall heat flux at midspan, and Tab. 11 reports the errors. The NASA values are reconstructed with Eq. (6). This is therefore not a comparison with an independent measurement, and it has a higher uncertainty than the previous ones.

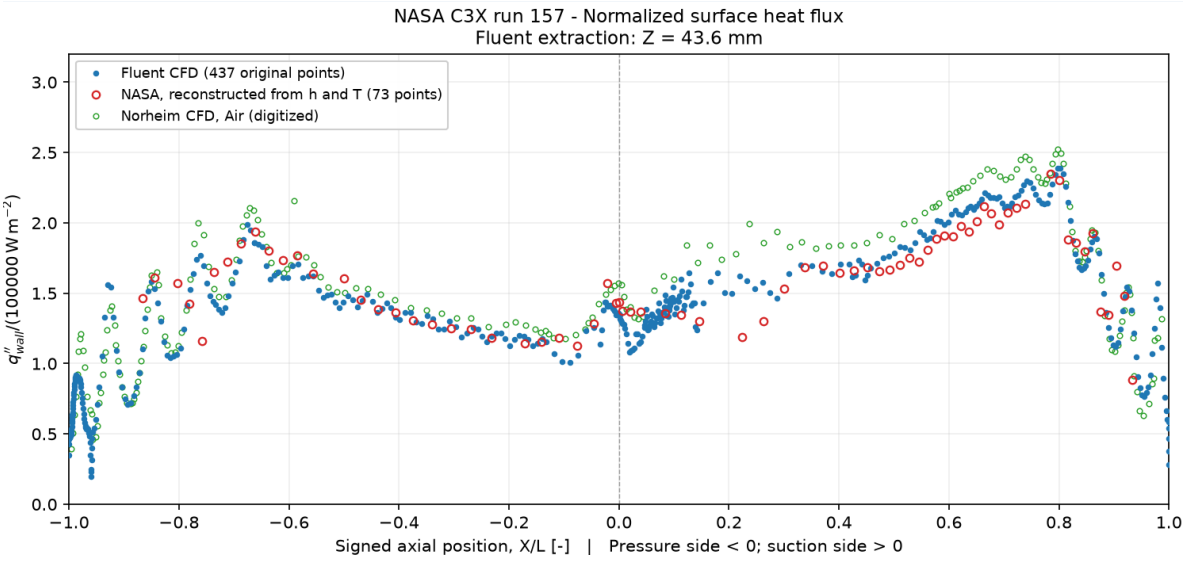


Figure 6: Normalised wall heat flux at midspan.

The heat flux has two peaks, at $X/L = -0.68$ on the pressure side and at $X/L = 0.80$ on the suction side. Both are at positions where the wall temperature has a local minimum (Fig. 5). There a cooling hole is close to the surface, the wall is colder, and more heat enters the vane. The position and the level of the two peaks are well predicted.

Table 11: Wall heat flux errors with respect to the NASA data.

Simulation	$\bar{e}_q$ [kW/m^2]	$\bar{e}_q$ [%]	$e_{q,\max}$ [kW/m^2]
Present work	13.1	8.78	56.8
Norheim and Amzin	20.6	13.40	77.6

For the present simulation the mean relative error is about 9%, which is comparable with the NASA uncertainty on the heat transfer coefficient. On average, the simulation is slightly below the reconstructed values on the pressure side, and slightly above them on the suction side. The maximum error $e_{q,\max}$ is in the rear part of the vane, where the uncertainty of the heat transfer coefficient is highest.

The most interesting region is the suction side between $X/L = 0.15$ and $X/L = 0.35$. Here the NASA heat transfer coefficient has a minimum at $X/L = 0.22$, and then it increases by about 50%.

NASA describes the behaviour of the C3X suction side as a transition of the boundary layer from laminar to turbulent [3]. The low-speed measurements of Ames on a larger model of the same profile also show a laminar boundary layer on the first part of the suction side, followed by transition [1].

The simulation does not show this feature. A plausible explanation is the absence of a transition model: the k–ω SST model treats the boundary layer as turbulent from the leading edge, so it cannot reproduce a laminar region. In this region the simulated heat flux is 20% to 30% higher than the reconstructed one, and $X/L = 0.22$ is also the station of the maximum temperature error $e_{T,\max}$ of Tab. 10. These facts are consistent with this explanation, but they do not prove it: no simulation with a transition model was run in this work.

The reference simulation shows the same behaviour in a stronger form. On average its heat flux is above the reconstructed values, in particular on the suction side, and its largest error on this side is at the same station, $X/L = 0.22$.

7.6 Final discussion

This section separates what is observed in the results, what is supported by the literature, and what remains a hypothesis.

Observations. The following points are directly supported by the results of Tabs. 9, 10 and 11:

- the present simulation reproduces the pressure within 2% and the wall temperature within 2.6% on average;
- the error on the heat flux is of the same order as the uncertainty of the NASA heat transfer coefficient;
- the simulated wall is hotter than the measured one at every station, by 15.7 K on average;
- the largest temperature error is on the front part of the suction side, where the simulated heat flux is also higher than the reconstructed one;
- with the same procedure, the digitised curves of the reference simulation give larger errors for all three quantities. This comparison is valid only for this run and for these digitised data. Because of the unresolved difference on the pressure error (Section 7.3), it is not a proof that the present simulation is more accurate.

Explanations supported by the literature. The local error on the suction side is in the region where the experiments report a laminar boundary layer and transition [3, 1]. Both reference studies relate their errors on the suction side to the turbulence model. Norheim and Amzin suggest a transition model as a possible improvement [5]. Froissart and Ochrymiuk show that the transition SST model gives a better shape of the temperature curve, but it underpredicts the temperature by about 20 K on average [2]. The absence of a transition model is therefore a plausible explanation of the local error, and not a simple one to fix. It was not tested in this work.

The literature also shows that the model options of Section 5 can change the wall temperature by 7 to 40 K [2]. These options are not reported in [5], so a part of the difference between the two simulations may come from them.

Hypotheses not verified in this work. The general offset cannot be attributed to a single cause with the available data. On the pressure side the wall is hotter than in the experiment, while the heat flux is on average slightly lower. An overprediction of the heat transfer on the gas side would raise both. This combination would therefore be consistent with a difference on the cooling side: a coolant that is too warm, a lower heat transfer inside the holes, or a lower conductivity of the steel. It is only an indication, because the difference in heat flux is within the NASA uncertainty.

Several inputs of the problem are not well known. The values below are approximate sensitivity estimates, obtained with a simple linear model. They are not results of sensitivity simulations.

- **Coolant inlet temperature.** A rough energy balance is consistent with the values used here. The heat that enters the vane, about 3.7 kW, warms the coolant by about 10 K between the inlet and midspan. This agrees with the difference between the NASA averages and the inlet values of the large holes. With the room-temperature assumption of [2], the coolant would be 40 to 60 K colder. The linear estimate indicates that a change of 10 K in the coolant temperature moves the wall temperature by 3 to 5 K.
- **Gas temperature.** With the same estimate, the NASA uncertainty of ± 11 K corresponds to 5 to 7 K on the wall temperature.
- **Conductivity of the steel.** The two reference studies use values that differ by about 15%. A lower conductivity, as used here, gives a hotter external surface.
- **Channel outlet pressure.** This value is a modelling assumption (Section 5.4). Its effect on the thermal solution is not known.

According to these estimates, the combined effect of the uncertain inputs is of the same order as the offset. Both the turbulence modelling and the uncertain boundary conditions therefore remain possible contributions, and neither is proved to be the cause.

8 Conclusions

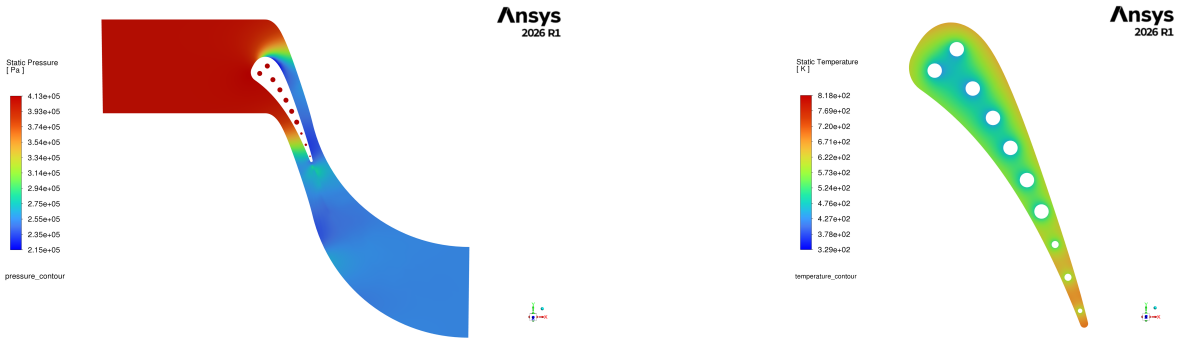
A complete CHT simulation of the NASA C3X vane was set up and run for run 157 with ANSYS ICEM CFD and ANSYS Fluent. A fully structured mesh of about 11 million cells was built for the external flow, the solid vane and the ten cooling channels, and the k–ω SST model was used. The main conclusions are the following.

- Mass and energy are conserved within 0.02% in all the fluid zones, and y^+ is below 1.5 on all the heat transfer walls.
- The results are in meaningful agreement with the NASA data. The mean relative error is 1.95% for the pressure and 2.55% for the wall temperature. For the reconstructed heat flux it is about 9%, which is of the same order as the experimental uncertainty.
- Some discrepancies remain. The wall temperature has the correct shape, but it is higher than the measured one at every station, by 15.7 K on average. The largest error, 34.6 K, is on the front part of the suction side.
- The absence of a transition model is a plausible explanation of the local error on the suction side, and the uncertain inputs, mainly the coolant inlet temperature, may explain a part of the general offset. Neither explanation was verified with dedicated simulations.
- Mesh independence and the sensitivity to the uncertain inputs were not verified. The results are therefore an assessment of one simulation, and not a complete validation of the model.

The project shows that the whole CHT workflow was implemented and checked, from the structured mesh to the comparison with the experiment. The main limits are the single mesh, the high continuity residual and the comparison limited to the midspan section. The natural next steps are a mesh independence study and sensitivity studies on the coolant inlet temperature, on the channel outlet pressure and on the turbulence model, including a transition model.

9 Appendix A: Flow-field visualization

Fig. 7 shows the contours of static pressure and of static temperature. They give a qualitative view of the results discussed in Section 7.



(a) Static pressure.

(b) Static temperature.

Figure 7: Pressure and temperature contours.

10 Appendix B: Vane coordinates

Tab. 12 lists the 78 points that define the 2D profile of the C3X vane [3]. The leading-edge radius is $R_{LE} = 1.168$ cm and the trailing-edge radius is $R_{TE} = 0.173$ cm.

Table 12: C3X vane coordinates.

n	x [cm]	y [cm]	n	x [cm]	y [cm]
1	0.1097	11.6548	40	7.4849	-0.0617
2	0.3894	12.1890	41	7.3188	0.3559
3	0.7658	12.6764	42	7.1483	0.7737
4	1.2723	13.0233	43	6.9736	1.1895
5	1.8743	13.1376	44	6.7950	1.6035
6	2.4707	12.9939	45	6.6116	2.0155
7	2.9835	12.6538	46	6.4237	2.4254
8	3.3985	12.1976	47	6.2309	2.8329
9	3.7376	11.6817	48	6.0328	3.2380
10	4.0272	11.1364	49	5.8296	3.6406
11	4.2885	10.5766	50	5.6203	4.0401
12	4.5326	10.0094	51	5.4051	4.4364
13	4.7648	9.4369	52	5.1834	4.8290
14	4.9870	8.8605	53	4.9548	5.2177
15	5.2019	8.2814	54	4.7191	5.6020
16	5.4110	7.7003	55	4.4760	5.9817
17	5.6157	7.1176	56	4.2248	6.3564
18	5.8171	6.5336	57	3.9654	6.7249
19	6.0160	5.9487	58	3.6975	7.0874
20	6.2126	5.3632	59	3.4204	7.4430
21	6.4074	4.7767	60	3.1339	7.7909
22	6.5997	4.1897	61	2.8374	8.1308
23	6.7894	3.6015	62	2.5314	8.4615
24	6.9756	3.0122	63	2.2149	8.7826
25	7.1575	2.4221	64	1.8885	9.0935
26	7.3335	1.8301	65	1.5519	9.3932
27	7.5024	1.2357	66	1.2052	9.6815
28	7.6624	0.6391	67	0.8494	9.9578
29	7.8115	0.0411	68	0.4999	10.2116
30	7.8161	-0.0053	69	0.3848	10.3035
31	7.8082	-0.0516	70	0.2822	10.4094
32	7.7879	-0.0935	71	0.1938	10.5273
33	7.7572	-0.1288	72	0.1212	10.6556
34	7.7180	-0.1542	73	0.0650	10.7920
35	7.6736	-0.1681	74	0.0264	10.9342
36	7.6269	-0.1699	75	0.0064	11.0802
37	7.5816	-0.1588	76	0.0046	11.2278
38	7.5408	-0.1356	77	0.0216	11.3741
39	7.5077	-0.1026	78	0.0569	11.5171

For point 29, the table reports the corrected coordinate $y = 0.0411$ cm. The original NASA table gives two values for this point, $y = 0.4115$ cm and $y = 0.0162$ in [3]. They do not agree, because 0.0162 in is equal to 0.0411 cm. The value in centimetres is therefore a typing error.

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